Distance Modulates Interference from Inappropriate QPs on Pronoun Resolution in Spanish

Background. How does linguistic structure interact with memory retrieval mechanisms during real-time sentence processing? Previous work [1-3] has shown that antecedent retrieval in pronoun resolution is robustly insensitive to interference from non-c-commanding QPs, such as *no girl scout* in (1). This is surprising because non-c-commanding QPs have been shown to cause facilitatory interference in NPI licensing [4,5], which likewise requires a QP to c-command a later NPI such as *ever* in (2). Crucially, [4] found interference only when the critical NPI was linearly close to the edge of the RC containing the QP, its scope domain (2), but not when it was more distant (3). They hypothesized that a representational update for the QP occurs at the edge of its scope domain, which eventually renders it "invisible" to retrieval. Abstracting away from NPI licensing, this view predicts that the closer something is to the locus of this update, the less likely the update has been completed and the greater the likelihood of interference. We hypothesize that the previously reported lack of interference in pronoun resolution in English is modulated by extraneous linear distance factors, rather than reflecting dependency-specific immunity. We test this hypothesis with Spanish object clitics, which can appear either pre-verbally (4) or post-verbally (5). Importantly, the distance between the edge of a QP's scope domain and the critical pronoun is shorter in pre-verbal than in post-verbal configurations. We exploit the possibility of clitic climbing in Spanish to test how distance, and thus the time, between a QP's scope domain and a later object pronoun in the main clause (MC) affects interference from inappropriate QPs. Across two self-paced reading (SPR) studies, we show that interference is greater in pre-verbal than in post-verbal configurations.

Design. We constructed 24 test items and 52 fillers. Items for both Experiment 1 and 2 followed a 2 x 2 design manipulating the factors *NP Type* and *Match*, see Table 1. Sentences started with a plural NP, to which a relative clause was attached. *NP Type* controlled whether the subject of the RC was *referential* (*el/la*) or *quantificational* (*ningún/a*). *Match* manipulated whether the NP *matched* or *mismatched* the critical object clitic (*lo*) in the MC. The object clitic was post-verbal in Experiment 1, and pre-verbal in Experiment 2. We used the gender-mismatch effect (GMME) [6,7] as an indicator of QP/NP accessibility to retrieval. If QP/NP was accessible, we expected matching pronouns to be read more quickly than mismatching pronouns.

Results (SPR). Average region-by-region log-converted RTs for both experiments can be seen in Figures 1 & 2. *Experiment 1, N=85***.** Linear mixed effects models revealed a significant Match * NP Type Interaction ($p < .05$) at the critical pronoun region. Planned comparisons showed a GMME in the *referential* conditions (diff = 40ms, $p < .01$), but not in the *quantificational* conditions (diff = 4ms, $p = 0.69$). In the spillover region, there was a main effect of Match ($p < .001$), qualified by a Match * NP Type Interaction ($p < .05$). Planned comparisons revealed a significant GMME in the *referential* conditions (diff = 66ms, $p < .001$), and not in the *quantificational* conditions (diff = 14ms, $p = .08$). *Experiment 2, N=86***.** There were no significant effects at the critical pronoun region. In the spillover region, only the main effect of Match was significant ($p < .001$). Planned comparison showed a GMME in both the *referential* (diff = 54ms; $p < .001$) and *quantificational* conditions (diff = 26ms; $p < .05$).

Discussion. Consistent with previous work, we found no GMME for quantificational sentences when pronouns were post-verbal, suggesting that matching non-c-commanding QPs do not significantly facilitate processing. In contrast, we observed a quantificational GMME for pre-verbal pronouns, consistent with an interference-based account. We propose that when a pronoun appears close to the edge of a QP's scope domain, the update that deactivates the QP as a potential antecedent may not yet be complete, leaving comprehenders susceptible to interference. Similar to NPI licensing, we show that the accessibility of a QP varies as a function of distance from the edge of its scope domain. Overall, our findings suggest that the previously reported lack of interference during antecedent retrieval for pronoun resolution follows from a dynamic update at some level of representation that renders the QP inaccessible to retrieval.

(1) The troop leaders $_{RC}$[that no girl scout had respect for] had scolded **her**$_{*i}$ after the incident.

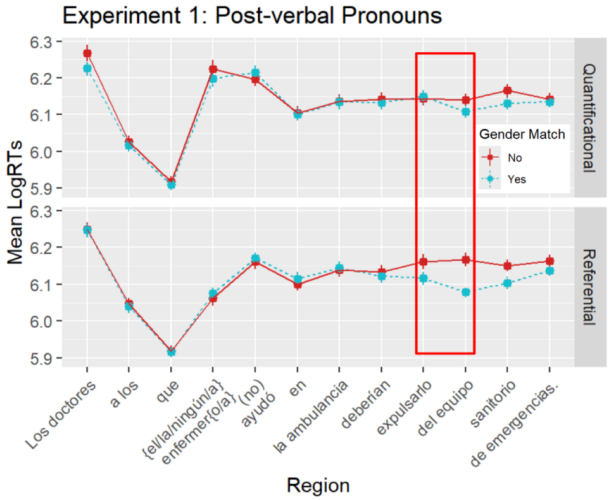
(2) *The authors $_{RC}$[that no critics recommended] **ever** received acknowledgment for a novel.
(3) *The authors $_{RC}$[that no critics recommended], as the editors mentioned, **ever** received acknowledgment for a novel.

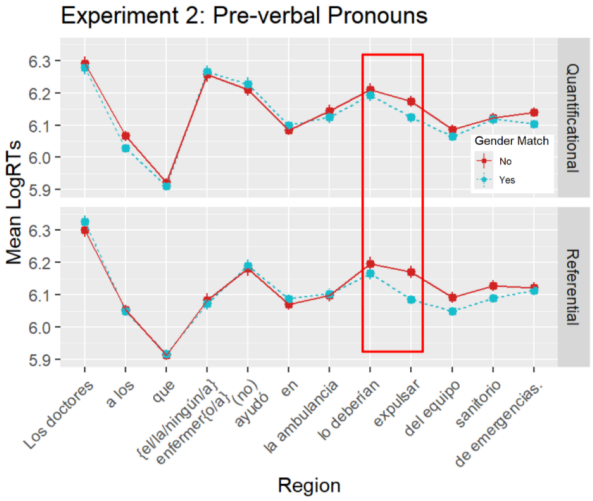
(4) Las profesoras $_{RC}$[a las que ningún niño$_i$ respeta] **lo**$_{*i}$ quieren castigar.
The teachers $_{RC}$[who no boy$_i$ respects] **him**$_{*i}$ want to punish.
(5) Las profesoras $_{RC}$[a las que ningún niño$_i$ respeta] quieren castigar**lo**$_{*i}$.
The teachers $_{RC}$[who no boy$_i$ respects] want to punish-**him**$_{*i}$.

Table 1. Sample set of experimental items for Experiment 1 (top) & Experiment 2 (bottom).

Experiment 1 (Post-verbal)	Referential	Los doctores $_{RC}$[a los que {el/la enfermero/a} no ayudó en la ambulancia] deberían expulsar**lo** del equipo sanitario de emergencia. *The doctors$_{RC}$[who {the nurse$_{-masc/-fem}$} did not help on the ambulance] should expel **him** from the emergency medical team.*
	Quantificational	Los doctores $_{RC}$[a los que {ningún/a enfermero/a} ayudó en la ambulancia] deberían expulsar**lo** del equipo sanitario de emergencia. *The doctors$_{RC}$[who {no nurse$_{-masc/-fem}$ helped on the ambulance] should expel **him** from the emergency medical team.*
Experiment 2 (Pre-verbal)	Referential	Los doctores $_{RC}$[a los que {el/la enfermero/a} no ayudó en la ambulancia] **lo** deberían expulsar del equipo sanitario de emergencia. *The doctors$_{RC}$[who {the nurse$_{-masc/-fem}$} did not help on the ambulance] **him** should expel from the emergency medical team.*
	Quantificational	Los doctores $_{RC}$[a los que {ningún/a enfermero/a} ayudó en la ambulancia] **lo** deberían expulsar del equipo sanitario de emergencia. *The doctors$_{RC}$[who {no nurse$_{-masc/-fem}$ helped on the ambulance] **him** should expel from the emergency medical team.*

Figures 1 & 2. Average log-converted RTs by region and condition for Experiment 1 (left) and Experiment 2 (right). The critical pronoun region is 'expulsarlo' in Experiment 1 and 'lo deberían' in Experiment 2.





References

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